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대규모 언어 모델 추론 엔진 비교: 최적화와 효율성을 중심으로

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Comparative Analysis of Large Language Model Inference Engines: Focusing on Optimization and Efficiency

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Abstract

Large language models (LLMs) are increasingly deployed in chatbots, code generation tools, search systems, and autonomous agents, making inference efficiency essential for real-world deployment. Recent workloads, such as chain-of-thought reasoning, self-consistency, test-time scaling, and agentic task execution, often require repeated model invocations for a single request, thereby increasing latency, memory usage, and serving cost. Although techniques such as batching, parallelization, quantization, caching, attention optimization, and model compression have been widely adopted, heterogeneous service requirements still make it challenging to select an appropriate inference engine.

This paper presents a systematic, service-oriented study of multiple open-source and commercial LLM inference engines. We evaluate their usability, deployment complexity, hardware support, scalability, and suitability for throughput-aware and latency-aware workloads. We further examine their design objectives by mapping each engine to its supported optimization techniques,

including batching, parallel execution, caching, compression, attention optimization, sampling acceleration, and structured output generation.

Based on this analysis, we identify key trade-offs in LLM inference infrastructure, including latency versus throughput, deployment scale, hardware diversity, and optimization flexibility. We also discuss future directions for inference engines, including complex LLM-based services, long-context inference, emerging transformer architectures, security, on-device execution, heterogeneous acceleration, and cloud orchestration. This study provides practical guidance for researchers and developers seeking to design efficient, scalable, and cost-effective LLM serving systems.

감사의글

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